

Yuzhe Zhang

Personal Information

Date of birth: January 10, 1998

Place of birth: Jiangsu, China

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Languages: Chinese (native), English (professional working proficiency)

Summary

I am a junior researcher focusing on **axionlike dark matter** detection using **nuclear magnetic resonance** and **quantum sensing** techniques. Within the Cosmic Axion Spin Precession Experiment (CASPER) collaboration, I have contributed to experiments, data analysis, and simulations, resulting in peer-reviewed publications and presentations at international conferences and seminars. Over the years, I have developed expertise in precision measurement, programming, and electronics. I also prioritize collaborative research, clear scientific communication, and teaching.

Education

Doctoral studies in Physics 11.2022–Present

Johannes Gutenberg University Mainz

Thesis: *Search for an Axionlike Dark Matter Candidate Using Nuclear Magnetic Resonance*

Master of Science, Physics 10.2020–10.2022

Johannes Gutenberg University Mainz

Thesis: *Search for Axion-like Particles by Performing NMR on a Thermally Polarized Sample*

Bachelor of Science, Physics 09.2016–06.2020

University of Science and Technology of China

Thesis: *Data Analysis in Earth Rotation Measurements Based on a Nuclear Spin–Electron Spin System*

Research

Cosmic Axion Spin Precession Experiment (CASPER)

10.2020–Present

GSI Helmholtz Institute Mainz & Johannes Gutenberg University Mainz

Advisor: Prof. Dr. Dmitry Budker

Contribution: Experiment, simulation, and data analysis. Design and optimization of axionlike dark matter detection strategy. Cross-team coordination within the CASPER collaboration.

Rb–Xe Comagnetometer Experiment

09.2019–06.2020

University of Science and Technology of China

Advisor: Prof. Dr. Dong Sheng

Contribution: Assistance in the experimental search for spin-gravity coupling. Analysis of co-magnetometer data.

Global Network of Optical Magnetometers for Exotic physics (GNOME)

06.2019–08.2019

Jagiellonian University

Advisor: Prof. Dr. Szymon Pustelny

Contribution: Simulation and data analysis for the GNOME collaboration.

Catalyst Research for Water Splitting

08.2017–06.2019

University of Science and Technology of China

Advisor: Prof. Dr. Jie Zeng

Contribution: Synthesized catalysts, prepared electrodes, and tested water-splitting performance.

Selected Publications

[1] J. Walter, O. Maliaka, Y. Zhang, et al., “Search for axionlike dark matter using liquid-state nuclear magnetic resonance,” *Phys. Rev. D*, vol. 112, p. 052 008, 5 Sep. 2025. DOI: 10.1103/39nc-vr9m

- Commissioning results of the CASPER-Gradient experiment.
- Co-first author (equal contribution; 3 authors). Contributions: apparatus construction and commissioning; data acquisition; analytical calculations and data analysis; manuscript preparation; coordination within the collaboration.

[2] Y. Zhang, D. A. Tumturk, H. Bekker, et al., “Frequency-Scanning Considerations in Axionlike Dark Matter Spin-Precession Experiments,” *Annalen der Physik*, vol. 536, no. 1, p. 2 300 223, 2024

- Theoretical considerations on frequency scanning for axion haloscopes.
- First and corresponding author. Contributions: all theoretical calculations and analysis; full manuscript preparation; coordination within the collaboration.

Presentations

Talks

2026 GNOME workshop

2023 Academic Seminar of Chinese Students and Scholars in the German–French Region

2022 Invited talk at a Seminar at Institute of Theoretical Physics, Chinese Academy of Sciences

2021 16th Patras Workshop on Axions, WIMPs and WISPs

Posters

2024 The Fundamentals School – summer school in high-energy and gravitational physics

2024 DPG Spring Meeting of the Atomic, Molecular, Quantum Optics and Photonics Section (SAMOP)

2023 The conference on dark matter, ions, molecules and atoms

2022 17th Patras Workshop on Axions, WIMPs and WISPs

Software Development

AxionBloch – [GitHub link]

- Simulation tool for axion-induced spin dynamics using Bloch equations.
- Contributions: all design and implementation.

Academic / Educational Contributions

2026	GNOME workshop Local co-organizer
2026	Educational publication – <i>Some Rules for Good Scientific Presentations</i> contribution: manuscript translation and editing
2025–2026	Teaching assistant in Advanced Physics Laboratory Course - Magnetic Resonance (undergraduate-to-master-level course)
2024	Educational publication – <i>Some rules of good scientific viewgraphs</i> idea initiation; manuscript writing and editing
2022	17th Patras Workshop on Axions, WIMPs and WISPs Local co-organizer

Skills and Experiences

Experiment / Technical

- **Nuclear Magnetic Resonance (NMR):** Apparatus construction, operation, calibration, and data acquisition.
- **Quantum sensing / precision measurements:** Operating Superconducting Quantum Interference Device (SQUID).
- **Cryogenics:** Managing liquid helium and nitrogen for experiments.
- **Hardware automation:** Programming-based control of laboratory equipment.

Programming / Computational

- **Data analysis:** Processing CASPER experimental data with Python; knowledge of particle physics data workflows from workshops.
- **Simulation:** Physics simulations (spin dynamics, signal modeling) using C++ and Python.
- **Numerical equation solving:** Solving Bloch and Schrödinger equations using C++ and Python.
- **Software tools:** Experience with Git / Docker / $\text{\LaTeX}$ / HDF5.
- **High-performance computing (HPC):** Conceptual understanding and practical usage learned through workshops.

Collaboration / Communication

- **International teamwork:** Collaboration with CASPER members. Hosting weekly data analysis meetings with international colleagues.
- **Mentoring and student support:** Assisting interns and students with laboratory integration and research projects; teaching assistant in university experiment course.
- **Scientific communication:** Co-authoring publications, poster presentations, and assisting in talks. Organization of international conference.